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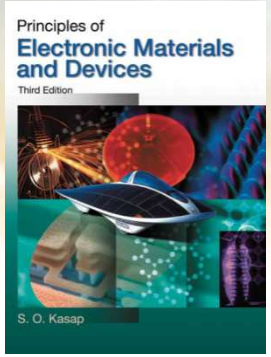
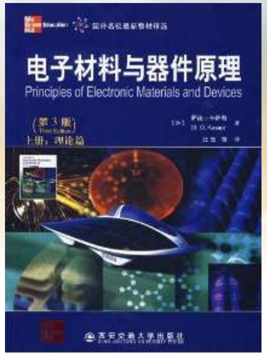
《固态电子学》
Solid-State Electronics





Ch1-3
Elementary Materials Science Concepts

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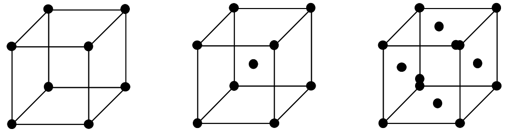
Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

立方晶系 CUBIC SYSTEM

$a = b = c \quad \alpha = \beta = \gamma = 90°$

Many metals, Al, Cu, Fe, Pb. Many ceramics and semiconductors, NaCl, CsCl, LiF, Si, GaAs

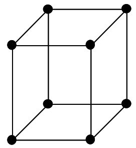


Simple cubic 简单立方 | **Body centered cubic** 体心立方 | **Face centered cubic** 面心立方

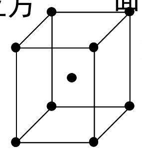
四方晶系 TETRAGONAL SYSTEM

$a = b \neq c \quad \alpha = \beta = \gamma = 90°$

In, Sn, Barium Titanate, TiO_2



Simple tetragonal 简单四方

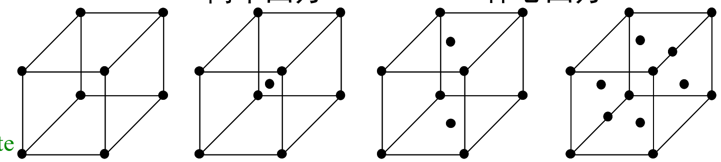


Body centered tetragonal 体心四方

正交晶系 ORTHORHOMBIC SYSTEM

$a \neq b \neq c \quad \alpha = \beta = \gamma = 90°$

S, U, Pl, Ga (<30°C), Iodine, Cementite (Fe_3C), Sodium Sulfate

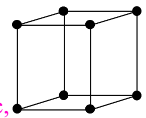


Simple orthorhombic 简单正交 | **Body centered orthorhombic** 体心正交 | **Base centered orthorhombic** 底心正交 | **Face centered orthorhombic** 面心正交

六方晶系 HEXAGONAL SYSTEM

$a = b \neq c \quad \alpha = \beta = 90°\ ;\ \gamma = 120°$

Cadmium, Magnesium, Zinc, Graphite

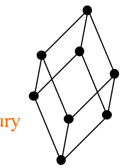


Hexagonal 六方

三方晶系 RHOMBOHEDRAL SYSTEM

$a = b = c \quad \alpha = \beta = \gamma \neq 90°$

Arsenic, Boron, Bismuth, Antimony, Mercury (<-39°C)

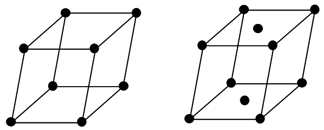


Rhombohedral 菱形

单斜晶系 MONOCLINIC SYSTEM

$a \neq b \neq c \quad \alpha = \beta = 90°\ ;\ \gamma \neq 90°$

α–Selenium, Phosphorus
Lithium Sulfate
Tin Fluoride

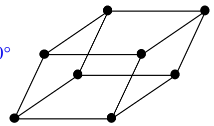


Simple monoclinic 简单单斜 | **Base centered monoclinic** 底心单斜

三斜晶系 TRICLINIC SYSTEM

$a \neq b \neq c \quad \alpha \neq \beta \neq \gamma \neq 90°$

Potassium dicromate

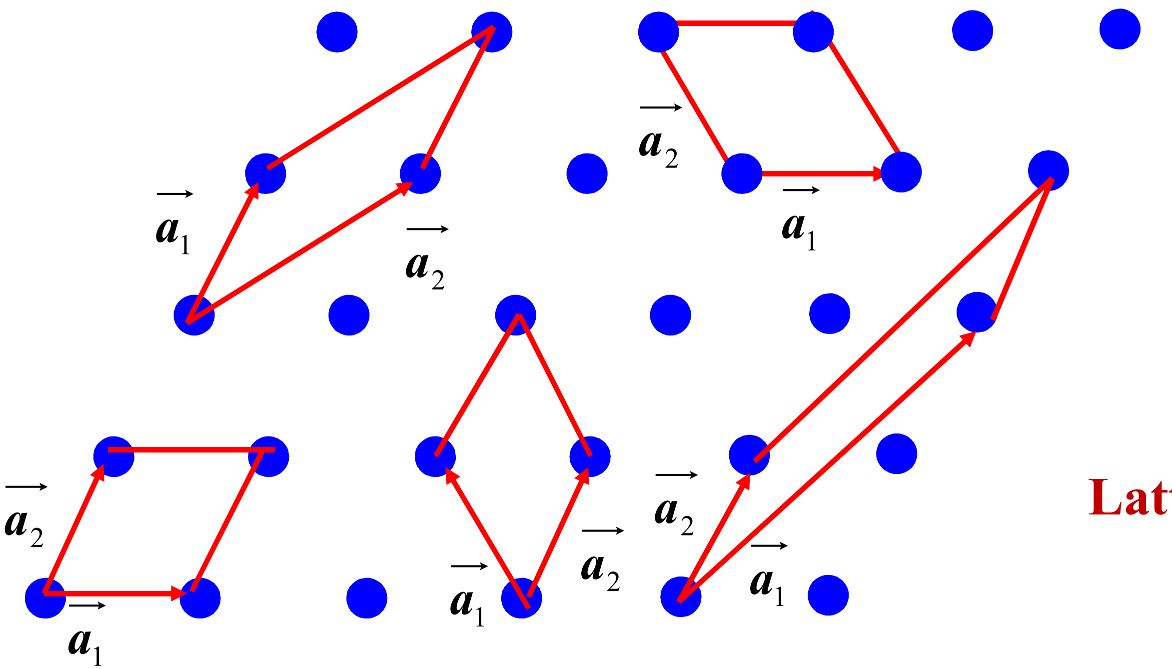


Triclinic 三斜

The seven crystal systems (unit cell geometries) and fourteen Bravais lattices.

The choice of unit cell has multiple possibilities

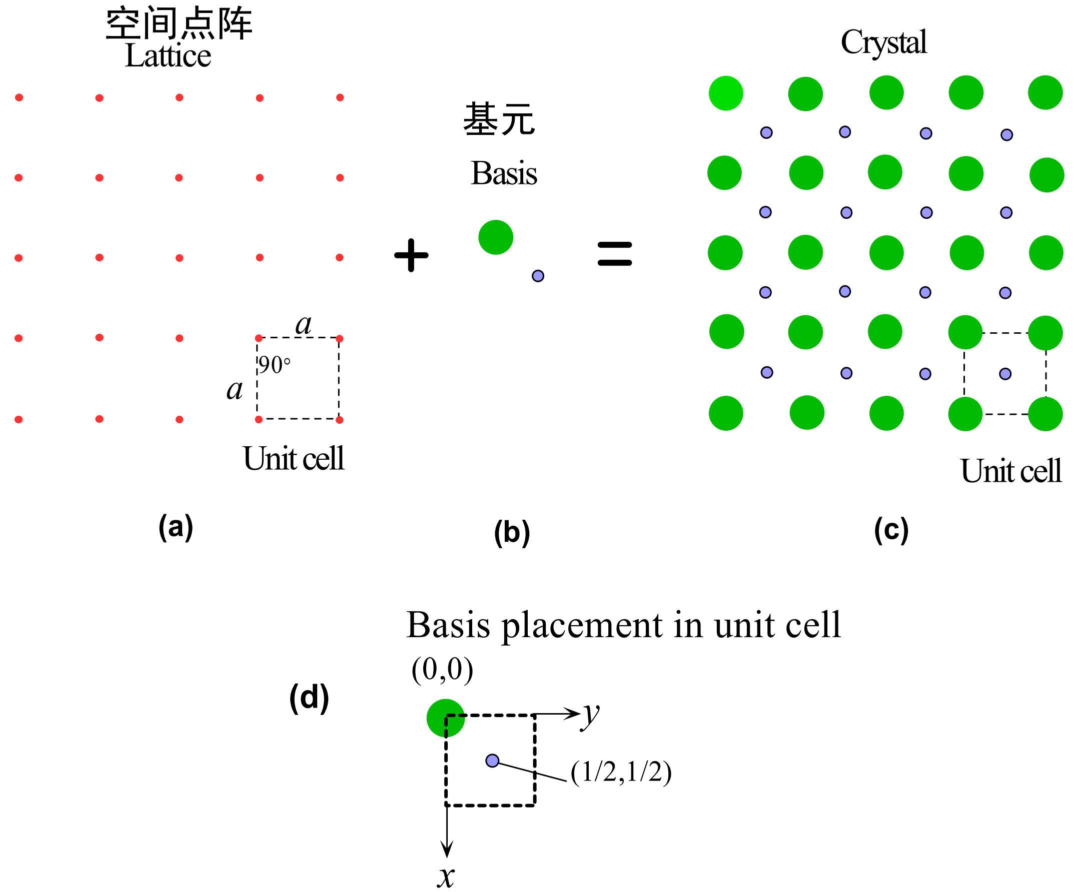
晶胞的选取不唯一



Lattice Parameters
晶胞参数

Characteristics: Periodicity, Symmetry, Minimum volume

Bravais Lattices 布拉维格子



(a) A simple square lattice. The unit cell is a square with a side *a*. (b) Basis has two atoms. (c) Crystal = Lattice + Basis. The unit cell is a simple square with two atoms. (d) Placement of basis atoms in the crystal unit cell.

An infinite periodic array of geometric points in space defines a space lattice（空间点阵） of simply a lattice（点阵）.

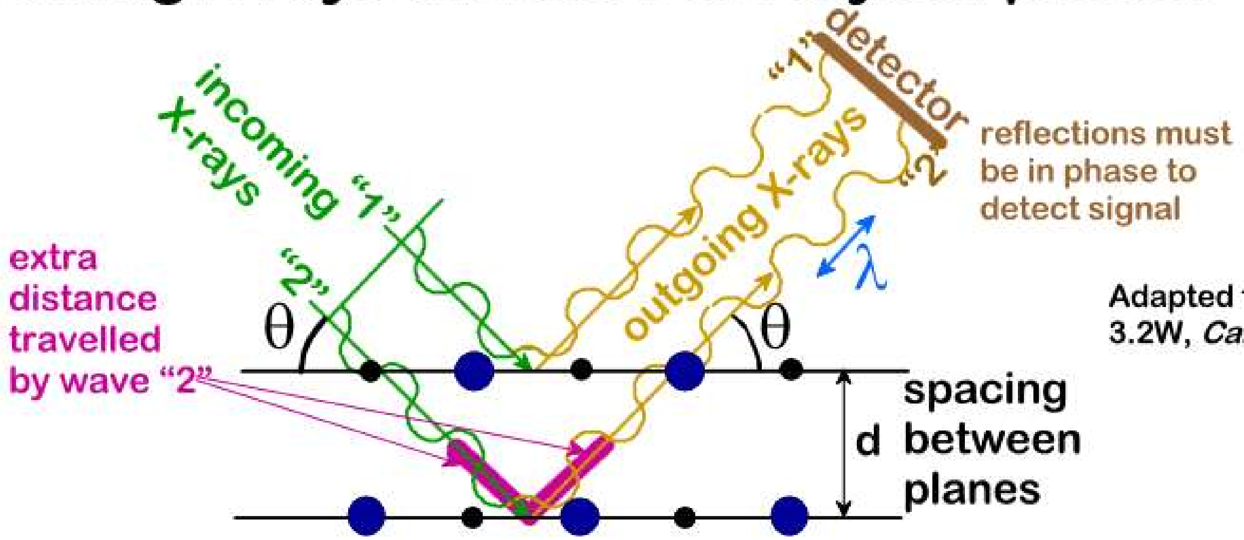
The lattice point -->
An identical group of atoms (Basis, 基元）

Crystal = Lattice +Basis

a: lattice parameter, 晶胞参数

X-RAYS TO CONFIRM CRYSTAL STRUCTURE

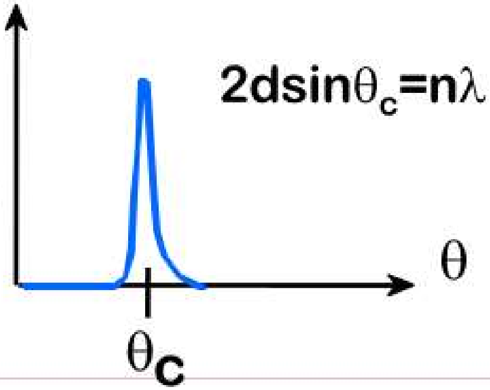
- Incoming X-rays diffract from crystal planes.



Adapted from Fig. 3.2W, *Callister 6e.*

- **Measurement of:**

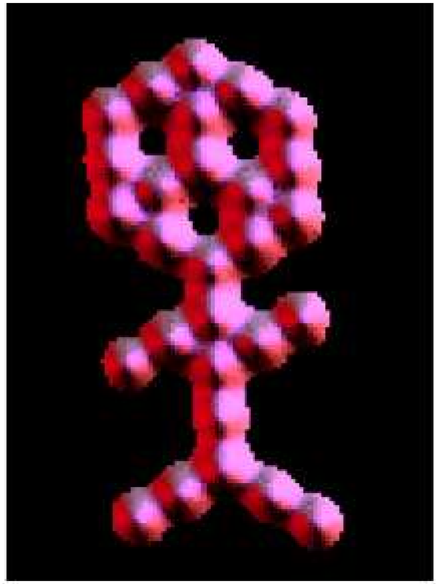
  Critical angles, θ_c, for X-rays provide atomic spacing, d.



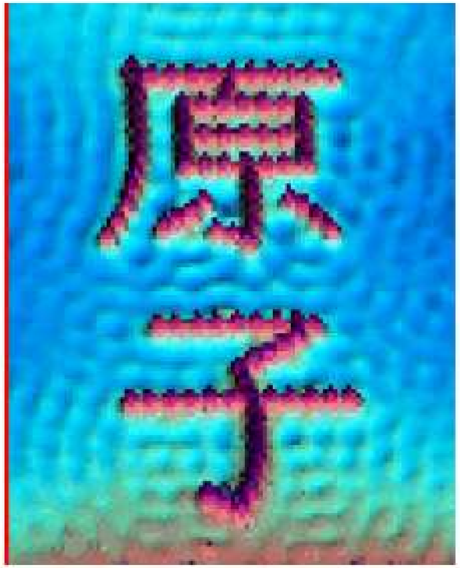
$$2d\sin\theta_c = n\lambda$$

SCANNING TUNNELING MICROSCOPY

- Atoms can be arranged and imaged!

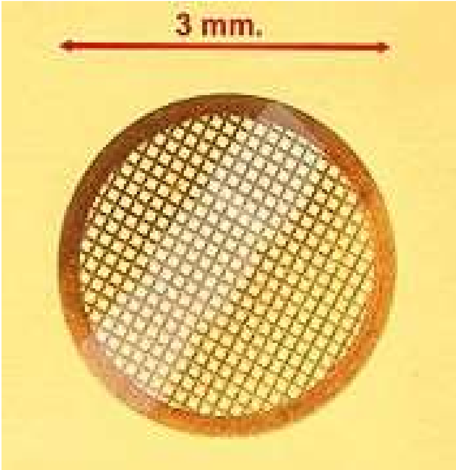


Carbon monoxide molecules arranged on a platinum (111) surface.



Iron atoms arranged on a copper (111) surface. These Kanji characters represent the word “atom”.

Transmission Electron Microscope (TEM)



$$d = \frac{\lambda}{2n\sin\alpha} \approx \frac{\lambda}{2\,\mathrm{NA}}$$

$$\lambda_e \approx \frac{h}{\sqrt{2m_0E\left(1+\frac{E}{2m_0c^2}\right)}}$$

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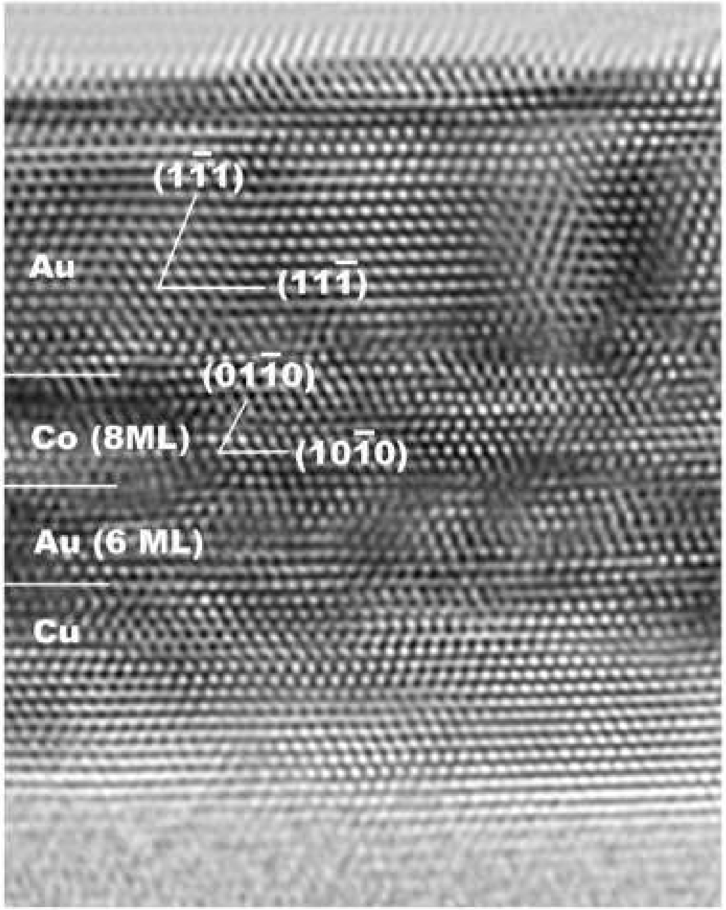
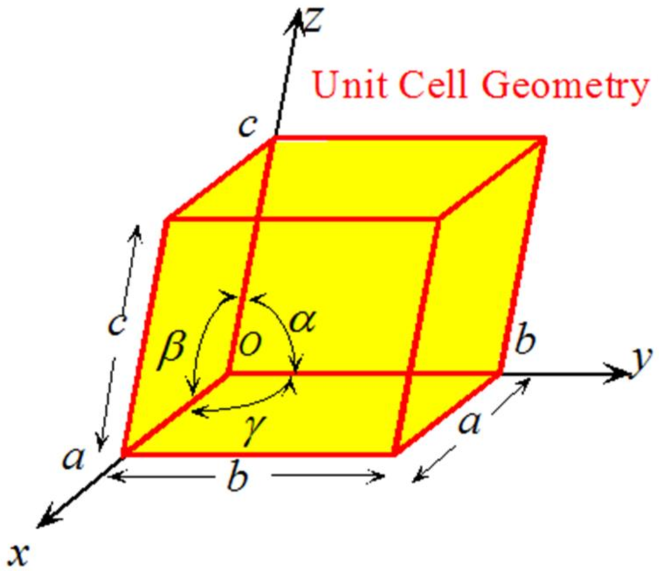
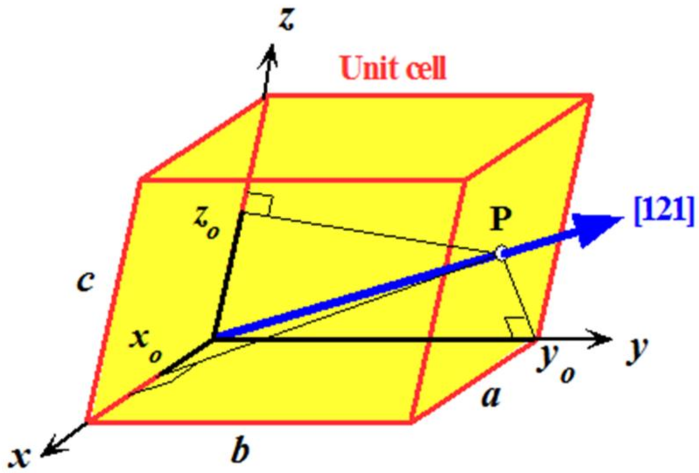
Imaging atoms with transmission electron microscopy (TEM)



1.8.2 Crystal directions and planes 晶向与晶面





Geometry of the unit cell

1）side（边）： *a, b, c*
2）angle（角度）： *α, β, γ*

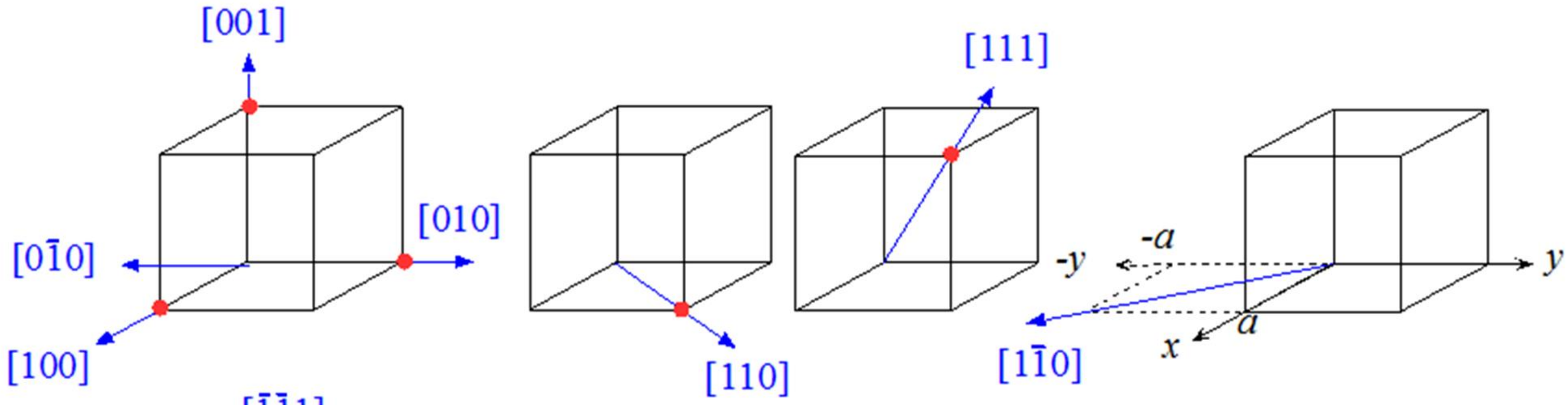
Any point can be expressed as: $l_1\vec{a} + l_2\vec{b} + l_3\vec{c}$

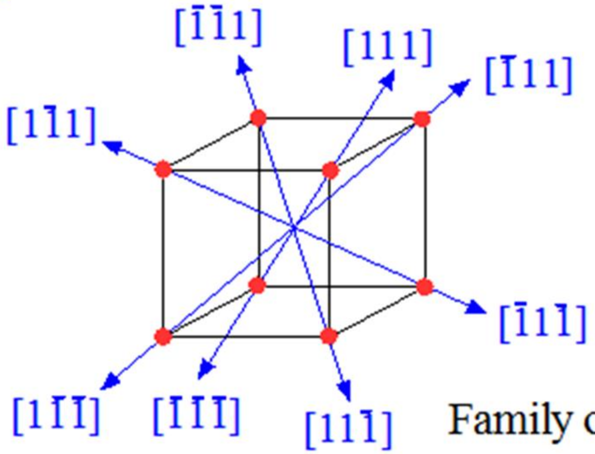
For example: $x_0, y_0, z_0 : \frac{1}{2}a, b, \frac{1}{2}c$

P is at: $x_p, y_p, z_p : \frac{1}{2}, 1, \frac{1}{2}$

Multiply or divide these numbers until we have the smallest integers (which may include 0), and the direction is written in square brackets without commas as $[1\,2\,1]$

Crystal directions 晶向





(c) Directions in cubic crystal system

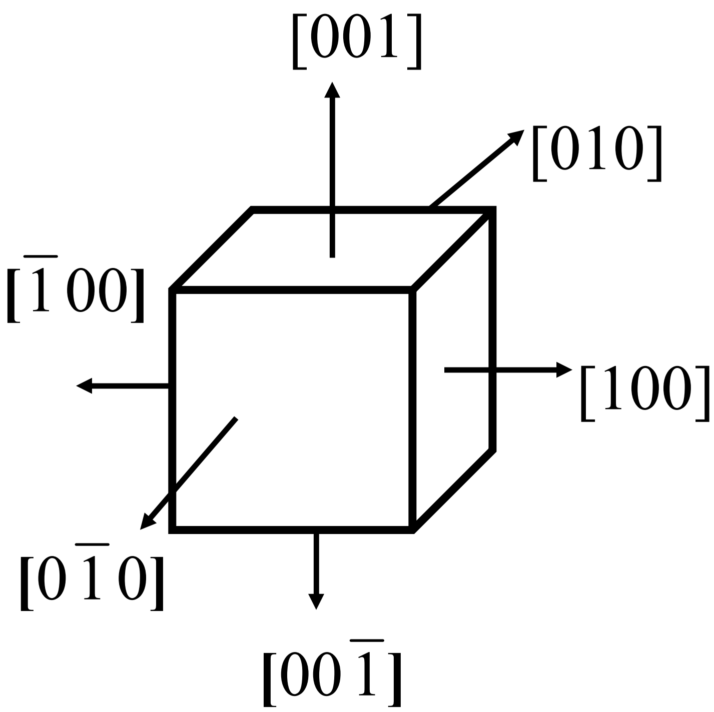
Family of <111> directions

Family of directions 晶向族

- Certain directions in the crystal are equivalent.

- For example, along the [010]and [001], all directional properties of a material are same. All of these directions constitute a family of directions（晶向族）。

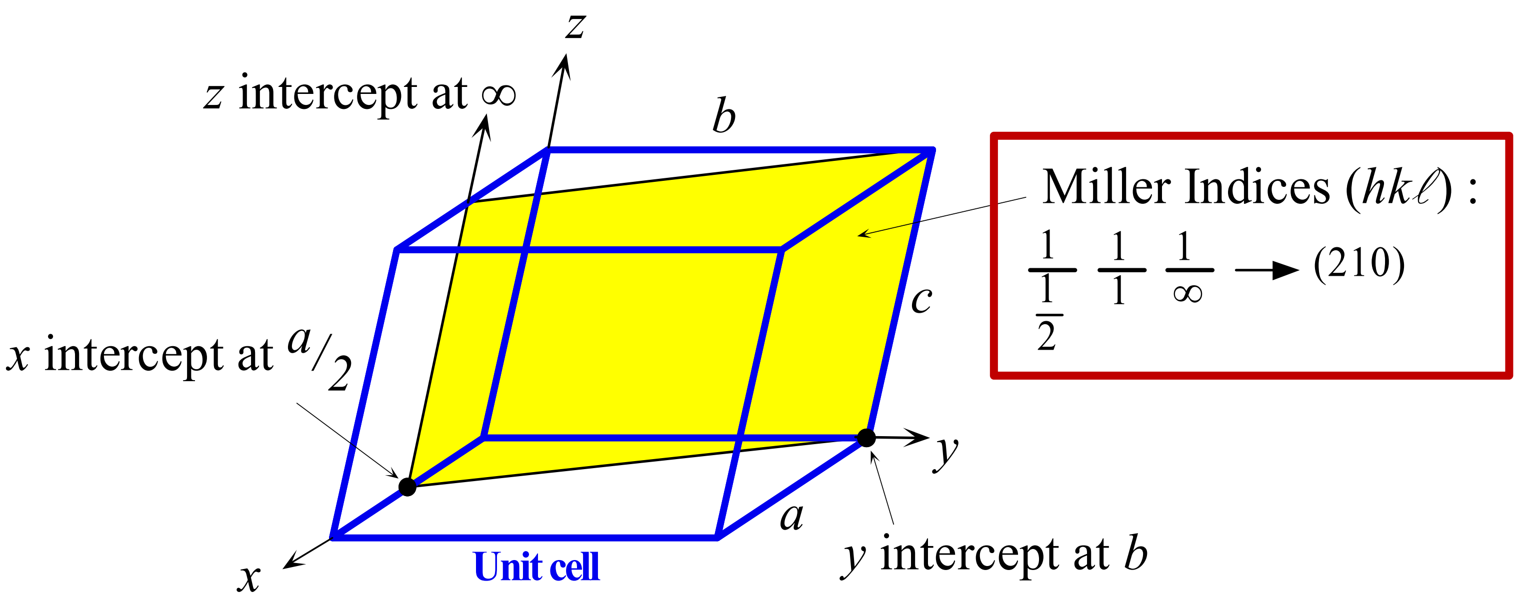
- Generally use triangle brackets < > to denote the family of direction



[100] and its equivalent directions

Family of direction: <100>

Miller indices of a plane 密勒指数



Intercepts x_o, y_o, and z_o are $\frac{1}{2}a$, $1b$, and $\infty\ c$.

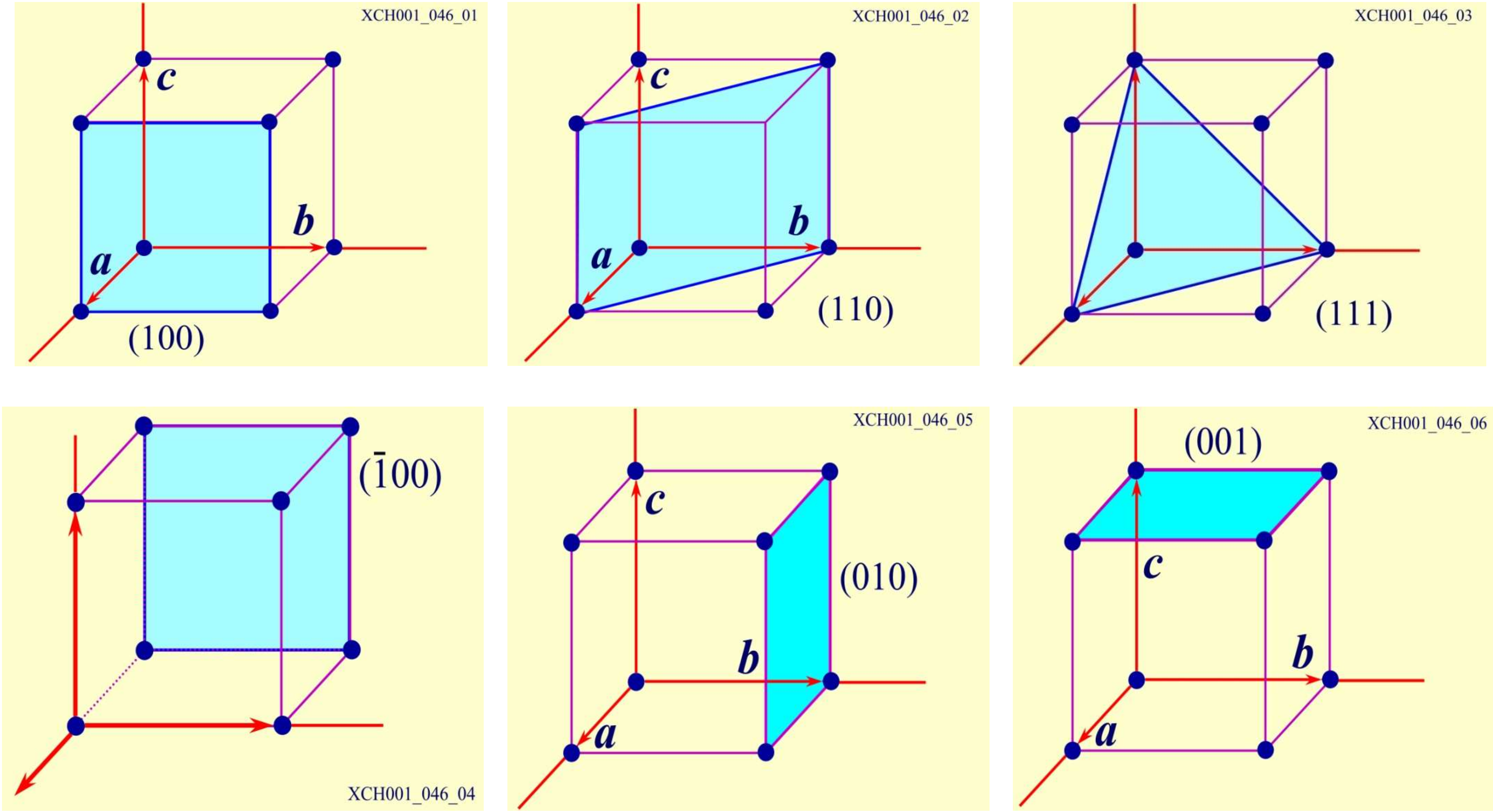
Intercepts x_1, y_1, and z_1, in terms of a, b, and c, are $\frac{1}{2}$, 1, and ∞.

Reciprocals $1/x_1$, $1/y_1$, and $1/z_1$ are $1/\frac{1}{2}$, $1/1$, $1/\infty = 2, 1, 0$.

密勒指数表述晶面，晶面在xyz轴的截距取倒数，再化简为整数。（括号）

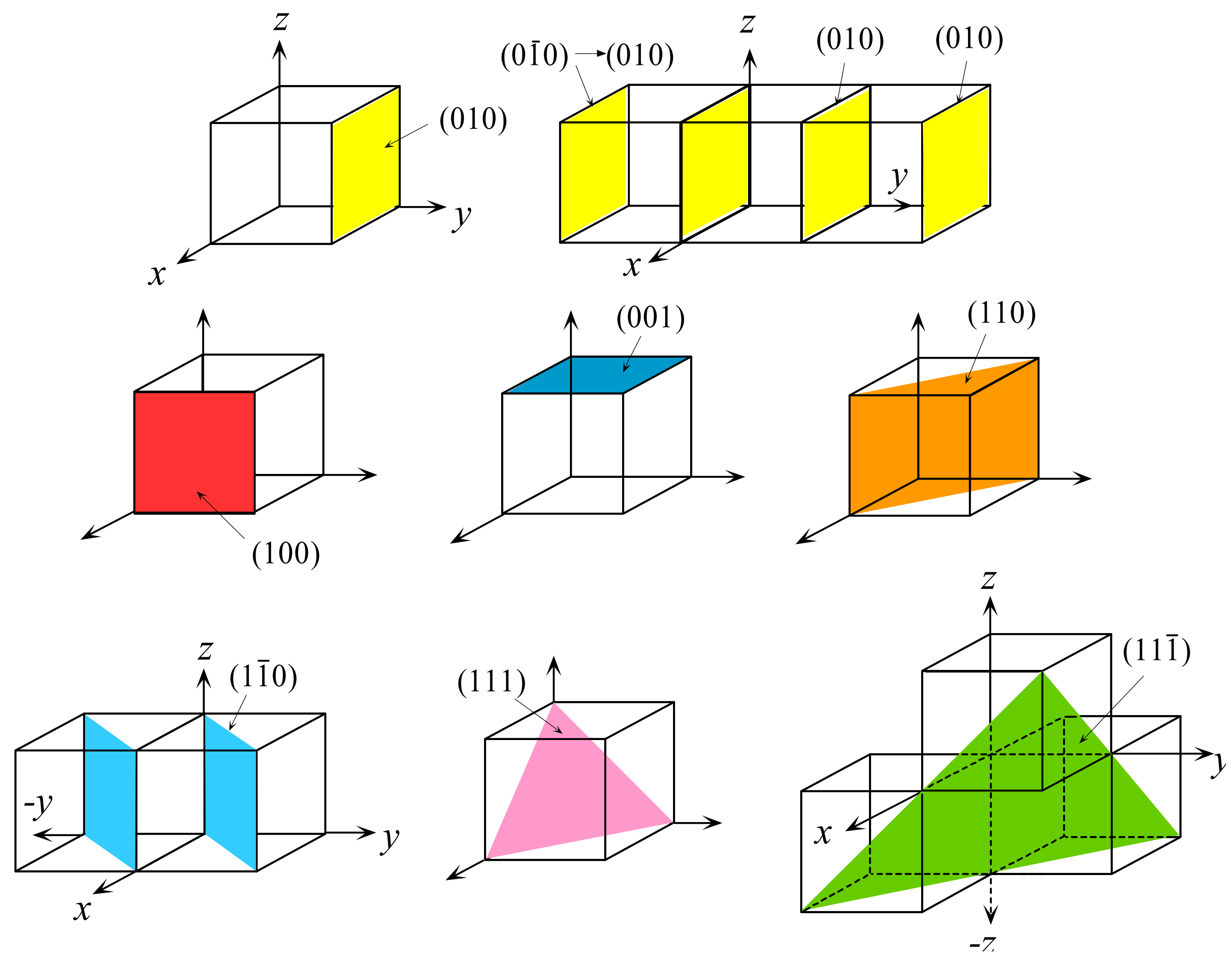
What is the relation between [*hkl*] and (*hkl*)?

Planes 晶面, Miller indices of a plane 密勒指数

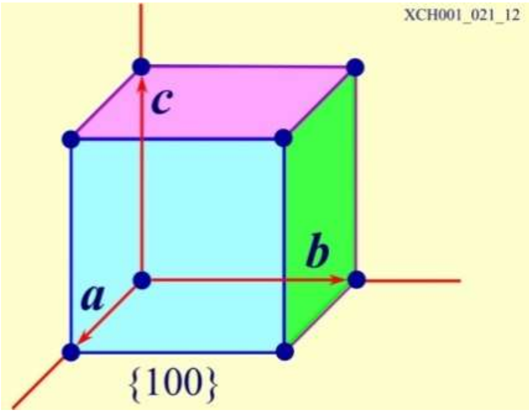


Indicate the position of plane (410)

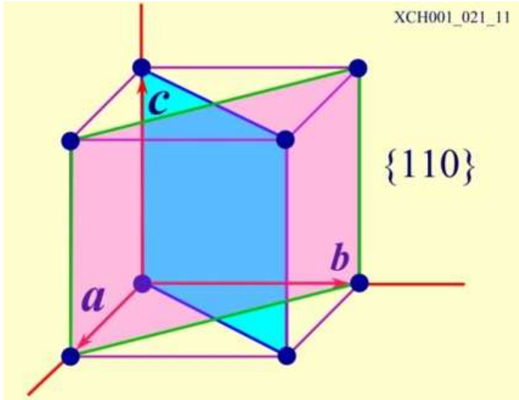
Planes 晶面



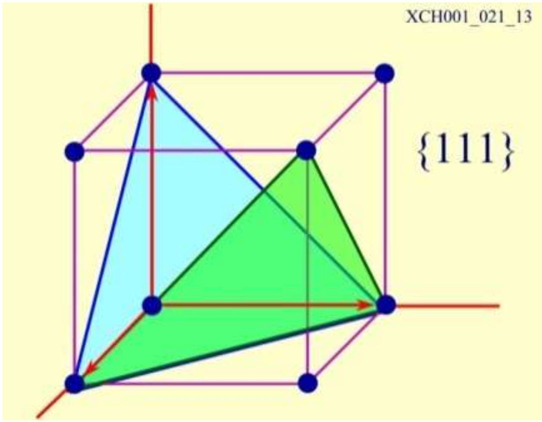
Family of Planes （晶面族）



(100)
(010)
(001)



(110) $(\bar{1}10)$
(101) $(10\bar{1})$
(011) $(01\bar{1})$

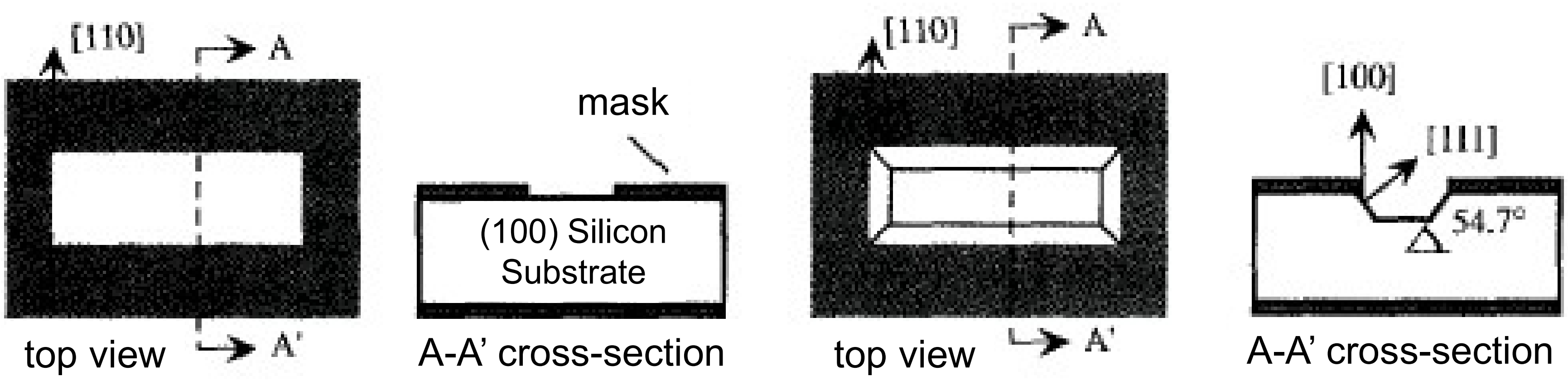


(111) $(11\bar{1})$
$(1\bar{1}1)$ $(\bar{1}11)$

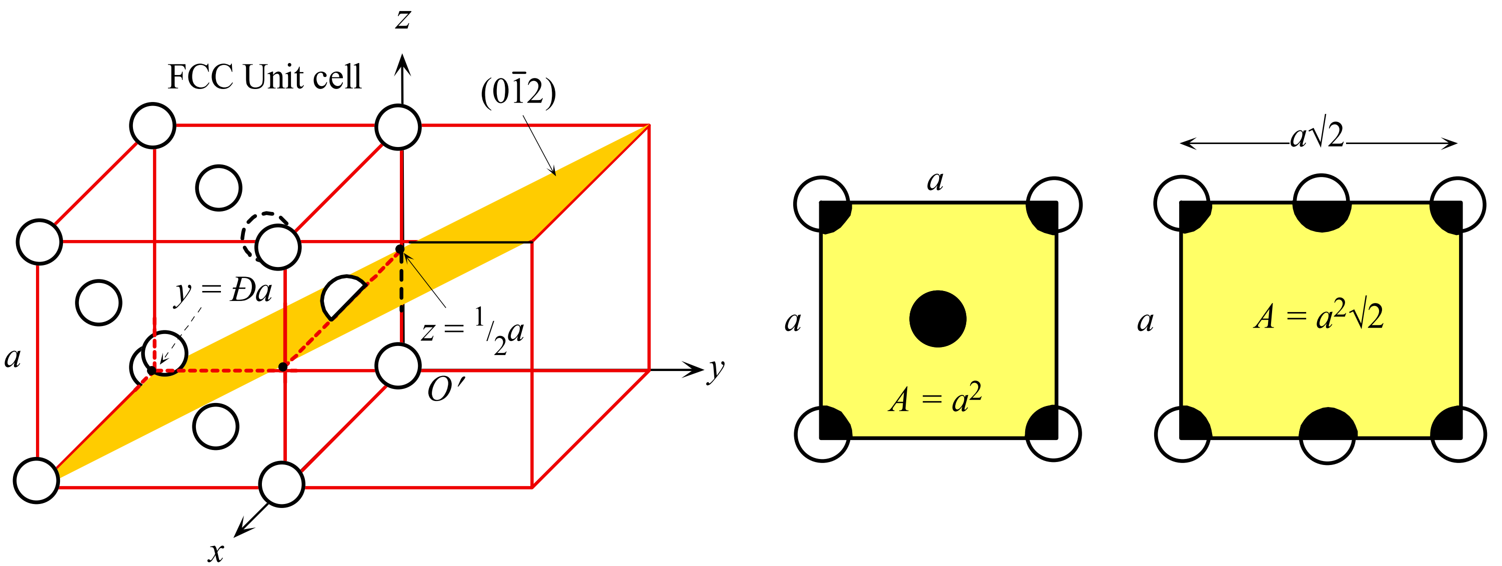
{ }family of planes, 晶面族

Crystal has anisotropic characteristics

- **Mechanical Property: cleavage, elastic modulus, etc.**
- **Thermal Property: thermal expansion coefficient, thermal conductivity…**
- **Electrical Property: conductivity**
- **Optical Property: refractive index**
- **Chemical properties: silicon wafers with (111) and (110) have different corrosion rates**



Planar concentration 晶面(原子)浓度



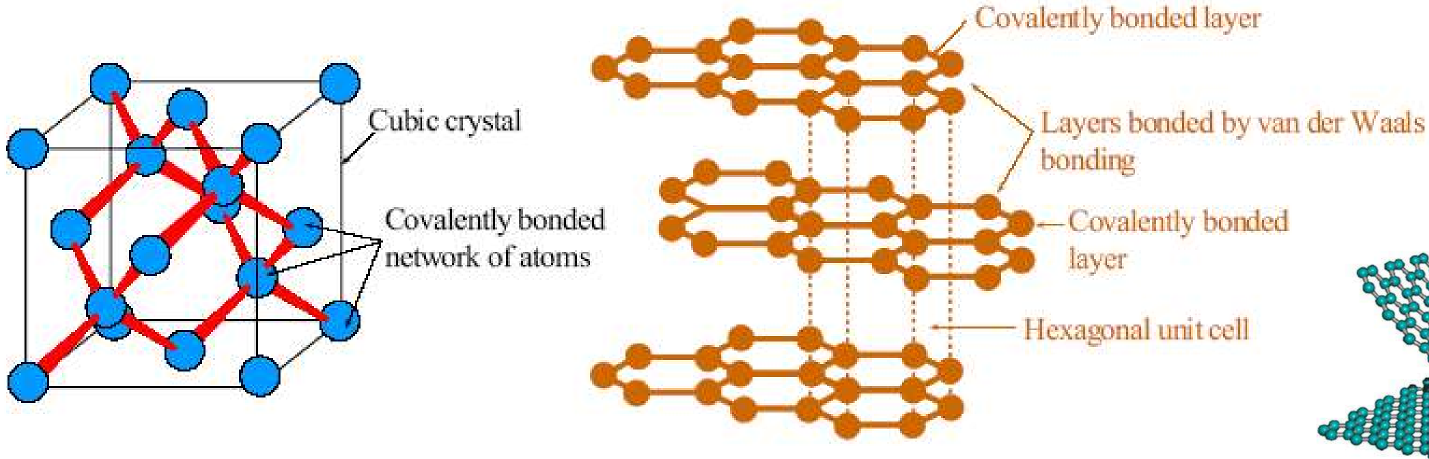
(a) (0$\bar{1}$2) **plane** **(b) (100) plane** **(c) (110) plane**

Calculate the planar concentration on (100), (110).

$$n_{(100)} = \frac{4(\frac{1}{4})+1}{a^2} = \frac{2}{a^2} = \frac{2}{(0.3620 \times 10^{-9}\ \mathrm{m})^2} = 15.3 \text{ atoms nm}^{-2}$$

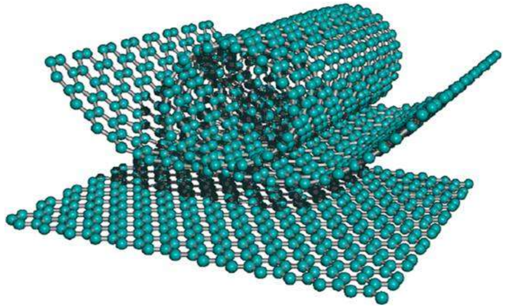
$$n_{(110)} = \frac{4(\frac{1}{4})+2(\frac{1}{2})}{(a)(a\sqrt{2})} = \frac{2}{a^2\sqrt{2}} = 10.8 \text{ atoms nm}^{-2}$$

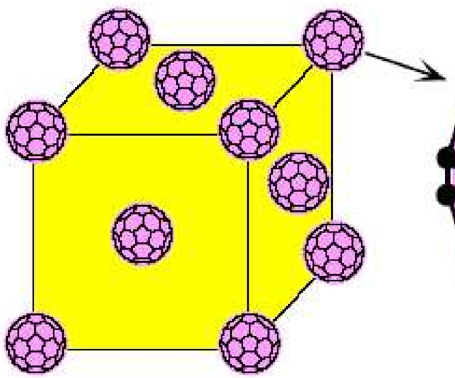
1.8.3 Allotropy and Carbon 同素异形体与碳



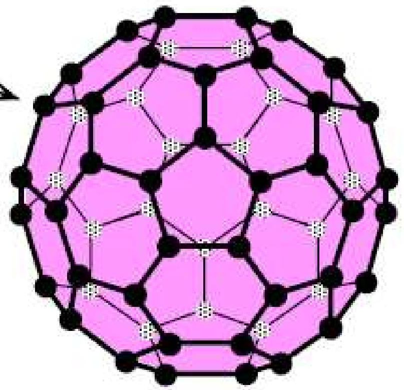
(a) Diamond unit cell

(b) Graphite





The FCC unit cell of the Buckminsterfullerene crystal. Each lattice point has a C_{60} molecule



Buckminsterfullerene (C_{60}) molecule (the "buckyball" molecule)

Polymorphism多形性 or allotropy同素异形: have more than one crystal structure.

1) < 912°C; BCC

2) 912<->1400°C; FCC

3) >1400°C; BCC

Transition temperature 相变温度: one allotrope changes to another at this temperature.

Table 1.4 Crystalline allotropes of carbon (ρ is the density and Y is the elastic modulus or Young's modulus)

	Graphite	Diamond	Buckminsterfullerene Crystal
Structure	Covalent bonding within layers. Van der Waals bonding between layers. Hexagonal unit cell.	Covalently bonded network. Diamond crystal structure.	Covalently bonded C_{60} spheroidal molecules held in an FCC crystal structure by van der Waals bonding.
Electrical and thermal properties	Good electrical conductor. Thermal conductivity comparable to metals.	Very good electrical insulator. Excellent thermal conductor, about five times more than silver or copper.	Semiconductor. Compounds with alkali metals (*e.g.*, K_3C_{60}) exhibit superconductivity.
Mechanical properties	Lubricating agent. Machinable. Bulk graphite: $Y \approx 27$ GPa $\rho = 2.25$ g cm^{-3}	The hardest material. $Y = 827$ GPa $\rho = 3.25$ g cm^{-3}	Mechanically soft. $Y \approx 18$ GPa $\rho = 1.65$ g cm^{-3}
Comment	Stable allotrope at atmospheric pressure	High-pressure allotrope.	Laboratory synthesized. Occurs in the soot of partial combustion.
Uses, potential uses	Metallurgical crucibles, welding electrodes, heating elements, electrical contacts, refractory applications.	Cutting tool applications. Diamond anvils. Diamond film coated drills, blades, bearings, etc. Jewelry. Heat conductor for ICs. Possible thin-film semiconductor devices, as the charge carrier mobilities are large.	Possible future semiconductor or superconductivity applications.